Automotive Grade Linux (AGL) Chiplet Abstraction Layer High level Architecture

1. Overview

The Automotive Grade Linux (AGL) Chiplet Abstraction Layer (CAL) is designed to enable modular, scalable, and hardware-agnostic compute architectures for Software-Defined Vehicles (SDVs). As the automotive industry shifts towards chiplet-based systems to meet performance and integration demands, CAL provides a unified interface to abstract heterogeneous chiplets (CPU, GPU, NPU, ISP, etc.) and expose their capabilities to the Linux operating system and user-space applications.

2. System Context

The CAL is integrated into the AGL stack as a kernel-level and user-space middleware component. It interfaces with the underlying chiplet-based hardware platform and provides standardized APIs and metadata to AGL's SoDeV (Software-Defined Vehicle) runtime. This enables seamless integration of multi-vendor chiplets and supports early hardware/software co-validation in virtualized environments.

3. Architecture Components

3.1 Chiplet Discovery Layer

Responsible for detecting and enumerating chiplets using firmware descriptors (e.g., ACPI tables, device tree overlays) and vendor-neutral metadata formats. It initializes chiplet entries in the system and registers them with the kernel.

3.2 Capability Abstraction Layer

Normalizes chiplet capabilities such as compute performance (e.g., TOPS, ISA), memory bandwidth, safety levels (ASIL), and power/thermal characteristics. Provides a consistent interface for querying chiplet features.

3.3 Interconnect Virtualization

Abstracts physical interconnects between chiplets, including UCIe, proprietary links, and high-speed memory channels. Enables dynamic routing and bandwidth allocation across chiplets.

3.4 Runtime Management

Manages chiplet lifecycle, including power states, thermal management, fault isolation, and OTA firmware updates. Integrates with AGL's existing power and thermal management frameworks.

3.5 User-Space Interfaces

Exposes chiplet information and control interfaces via sysfs, /dev nodes, and gRPC/DBus services. Enables integration with AGL middleware and containerized applications.

4. Data Flow

1. Firmware initializes chiplet descriptors during boot.
2. Chiplet Discovery Layer parses descriptors and registers chiplets.
3. Capability Abstraction Layer queries and normalizes chiplet features.
4. Interconnect Virtualization maps chiplet communication paths.
5. Runtime Management monitors and controls chiplet states.
6. User-space interfaces expose chiplet data to AGL services and applications.

5. Integration Points

- AGL SoDeV: CAL integrates with SoDeV to support hardware abstraction and dynamic configuration.
- Kernel Subsystems: Interfaces with device tree, ACPI, power management, and thermal frameworks.
- User-Space Tools: chipletctl CLI, libchiplet API, and gRPC services for orchestration and diagnostics.

6. Security and Safety Considerations

- Isolation: Ensures chiplet-level fault containment and memory protection.
- ASIL Compliance: Supports safety-critical workloads with deterministic behavior.
- Firmware Integrity: Verifies chiplet firmware signatures and supports secure OTA updates.

7. Roadmap

1. Define chiplet metadata schema aligned with CHASSIS and imec standards.
2. Develop kernel-level discovery and capability abstraction drivers.
3. Implement user-space interfaces and integration with AGL SoDeV.
4. Create virtual chiplet simulators for early development.
5. Validate on hybrid and physical platforms.
6. Submit upstream proposals to AGL expert groups.